

MYELIN REPORT

RESEARCH REPORT

A Technical Report on Behavioral Monitoring, Intervention and Recovery with MYELIN

GENERATED BY MYELIN CORE

guide . recover . focus

From Intent to Drift, Absorption, and Re-engagement

ABSTRACT

Digital work is frequently interrupted by application switching, messaging, social platforms, streaming media and other competing activities. Conventional screen-time measures describe where time was spent, but they reveal much less about what happened to the task that preceded the interruption or how behaviour recovered afterwards.

Keywords: digital interruption, task switching, behavioural return, re-engagement, digital distraction, human-computer interaction

1 | INTRODUCTION

A person may begin a digital task with a genuine intention to work, maintain that task for a period, transition to another activity, remain engaged elsewhere and eventually return. The return, however, should not automatically be interpreted as restoration of the state that existed before the interruption.

Drift, competing tasks and return

The neuroscience of drift requires an important distinction between internal mind-wandering and external attentional redirection. Mind-wandering research implicates both default-mode and executive-system contributions rather than a simple switch from a focus network to a distraction network.

MYELIN measures the behavioural geometry around attention -- where activity goes, how long it remains elsewhere and how successfully it returns -- not the hidden biology generating that behaviour.



Figure 1 | Conceptual behavioural trajectory studied by MYELIN

Scope note:
The sequence describes the observable behavioural trajectory surrounding intended digital work. The terms do not imply that MYELIN directly measures attention, motivation, neural activity or cognitive state. The chapters that follow trace this trajectory through each MYELIN organ, the certified evidence each stage produces, and the results and conclusions that evidence can support.

2 | DRIFT INTERVENTION ORGANS: SINE AND MINE

SINE and MINE are MYELIN's two primary drift-intervention organs. Both regulate off-task digital behaviour, but MINE introduces a branch pathway for messaging the user considers important.

SINE

SOFT INTERVENTION NUDGE ESCALATION

SINE is MYELIN's general drift-regulation organ. It responds when software-observed behaviour suggests that the user has moved away from the intended task context into a distracting digital pathway. It uses staged escalation that begins softly and becomes progressively firmer if off-task behaviour persists.

Core purpose:
regulates general digital drift through progressive intervention.

Distinctive features:
Blue to Amber to Protected Red escalation; return is not proven by a button press alone; MIRL governs recovery verification.

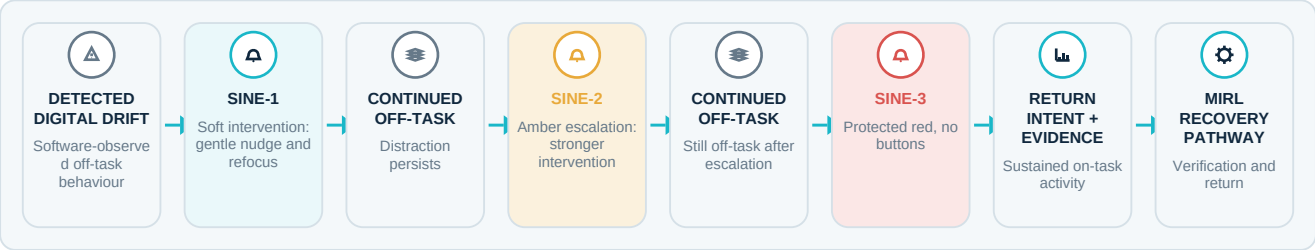


Figure 2 | SINE workflow

SINE is an escalation organ: intervention intensity increases as off-task exposure persists. The green Normal Break welcome-back screen is not an additional SINE stage.

MINE

MESSAGING INTERVENTION NUDGE ESCALATION

MINE is MYELIN's messaging-specific organ. It activates when qualifying messaging behaviour persists long enough to suggest that conversation activity is taking control of the session. Unlike SINE, MINE contains a branch structure that distinguishes ordinary messaging drift from communication the user considers important.

Core purpose:
regulates persistent messaging behaviour through branch-based escalation.

Distinctive features:
MINE-1 presents two options; both later converge to a final no-button callback; MINE owns its own recovery sequence.

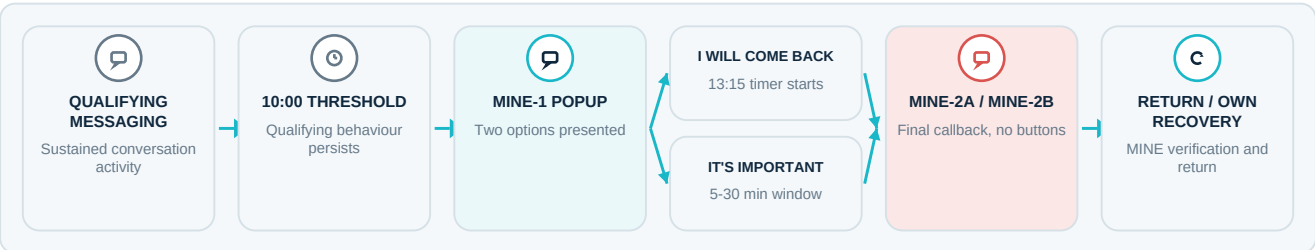


Figure 3 | MINE workflow

Messaging-app switches remain within the same episode; neutral detours pause the post-MINE-1 timer rather than reset it.

2 | CONTEXT-SPECIFIC REGULATION ORGANS: COSINE AND COSECANT

COSINE and COSECANT regulate specialised contexts within MYELIN. COSINE manages OTT streaming pathways, whereas COSECANT governs return from an approved Study break to the exact registered learning resource.

COSINE

CHOICE-ORIENTED STREAMING INSIGHT AND NAVIGATION ENGINE

COSINE is MYELIN's OTT and streaming-regulation organ. It is designed for prolonged entertainment-streaming contexts rather than ordinary web drift or messaging. COSINE first asks about the user's intention, then manages either return intention or an intentionally chosen leisure interval before guiding the user back toward task resumption.

Core purpose:
regulates OTT and video diversion through intention-aware guidance.

Distinctive features:
Separates return intention from intentional leisure viewing; both pathways eventually converge to a callback and return route.

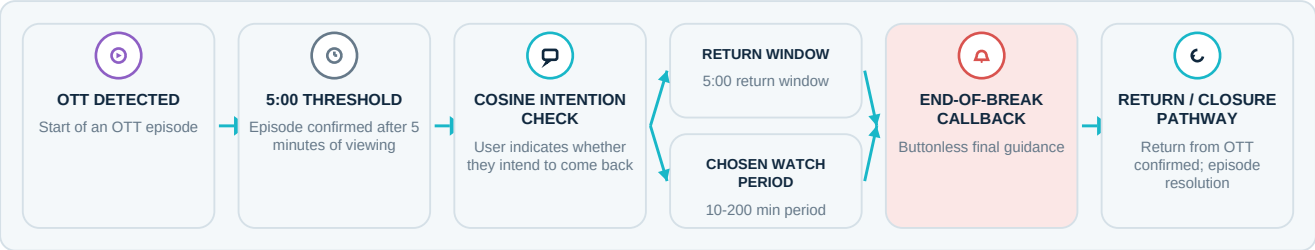


Figure 4 | COSINE workflow

OTT platform switches remain inside the same COSINE episode. The current production pathway does not use the older '10 more min' / 'finish my show' model.

COSECANT

CONTEXT-ORIENTED STUDY ESCALATION, CONFIRMATION AND NAVIGATION TOOL

COSECANT is the return-regulation organ of Study Mode. Study Mode begins by defining study time and task-relevant keywords or resources. When the user remains on the exact validated study resource, SINE may be withheld; movement to an unrelated resource can restore SINE eligibility. If the user takes an approved break, COSECANT manages the route back to the exact registered study resource.

Core purpose:
protects return-to-study continuity after an approved break.

Distinctive features:
Works from a finite break allowance; uses break-complete to +5 min to +1 min escalation; requires exact-resource confirmation rather than intention alone.

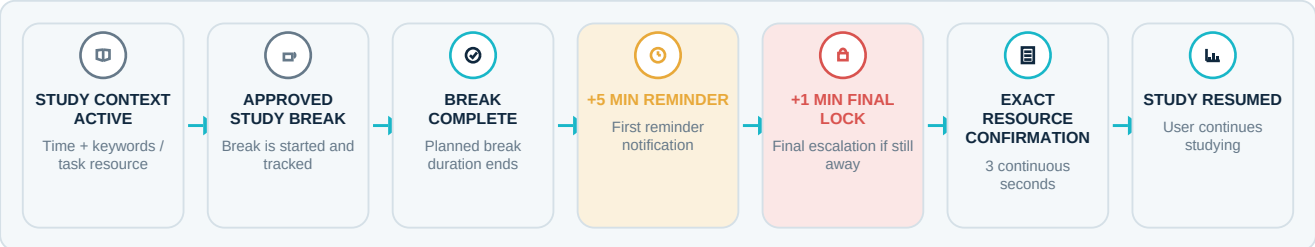


Figure 5 | COSECANT workflow

COSECANT can be understood as the opposite boundary of SINE: SINE escalates during off-task drift, whereas COSECANT begins from a permitted break and governs the countdown back to study continuity.

3 | WHAT DATA MYELIN ACTUALLY COLLECTS

This chapter explains the behavioural data MYELIN can record from certified software evidence. It describes what the system actually collects, not live results or invented numbers.

HOW MYELIN COLLECTS DATA

MYELIN records behavioural evidence from certified software when you work on a task. It captures activity related to task work, drift, intervention, return, and recovery across the different MYELIN organs. Report values update when new eligible sessions are added and a new PDF is generated.

KEY BEHAVIOURAL METRICS

Science Lab defines the following behavioural metric categories. Each metric is reported only when certified evidence supports it; metrics with no certified read path under the current Chapter 3 contract are marked unavailable rather than shown as if already measurable.

Task-aligned time

Time spent on the intended task. Not yet supported by the certified Chapter 3 contract.

Off-task dwell

Time spent away from the intended task. Not yet supported by the certified Chapter 3 contract.

Return latency

How long it takes to come back after drift. Reported when certified MIRL evidence exists.

Verified returns

Confirmed returns supported by evidence. Reported when certified MIRL evidence exists.

Relapses

Returns that were followed by drifting away again. Reported when certified MIRL evidence exists.

Post-return persistence

How long on-task behaviour continues after return. Not yet supported by the certified Chapter 3 contract.

Context switches

How often the user changes between task contexts. Not yet supported by the certified Chapter 3 contract.

Experiment duration

The total measured time for the selected experiment or session. Always available once an experiment is selected.

LIVE EVIDENCE FEED

Chronological behavioural events are recorded here from certified software evidence.

> Examples include intervention display, off-task activity, return attempts, verified return, relapse, and closure-related events.

DERIVED FINDINGS

This area turns validated evidence into comparisons and summaries.

> Examples include best and worst return times, common drift patterns, and organ-specific behaviour.

SOURCE COVERAGE

MYELIN can collect evidence from the following organs and sources when available. Each can contribute different types of behavioural data.

SINE
Intervention evidence

MINE
Messaging evidence

COSINE
OTT evidence

MIRL
Recovery evidence

COSECANT
Study-break return evidence

RESOURCE INTENT
Task-context evidence

STUDY
Study-context evidence

NORMAL BREAK
Break evidence

IMPORTANT BOUNDARY

MYELIN measures software-observed behaviour. It does not directly measure dopamine, brain activity, motivation, boredom, stress, fatigue, or subjective effort.

3 | HOW MYELIN USES THE COLLECTED DATA

This page explains what MYELIN can learn from its collected evidence. It focuses on behavioural interpretation, HCI relevance, and careful prediction rather than direct brain measurement.

FROM EVIDENCE TO INTERPRETATION

MYELIN first records software-observed behavioural events from certified sessions, such as task engagement, drift, interventions, returns, and context switches. These events are then organised into behavioural metrics, validated comparisons, and session summaries.

WHAT THE EVIDENCE CAN SHOW

The collected evidence can provide insights into several aspects of user behaviour and system performance.

Return latency

How quickly a user comes back after drift or an interruption.

Platforms involved

Which apps, sites, or organ pathways were involved in drift and return.

User reaction

What option the user chose or how they responded to an intervention pathway.

HCI impact

How the evidence can help evaluate prompt timing, burden, escalation, and design effectiveness.

Neuroscience prediction

May suggest behavioural hypotheses about re-engagement or difficulty of return, but this is a prediction from behaviour, not a direct brain measurement.

FROM SESSION EVIDENCE TO FINDINGS

MYELIN turns raw session events into structured findings through a series of steps.

EVIDENCE EVENTS
Off-task activity, Verified return, Relapse

→

METRICS
Durations, Counts, Frequencies

→

PATTERNS
Organ and platform trends, Common sequences

→

INTERPRETATION
HCI insight, Behavioural explanation, Research questions

WHOLE-MYELIN ANALYSIS

Once a full MYELIN run is complete, the system can combine evidence from all organs into a single session-level summary. This supports both practical design insights and scientific analysis.

LIMITATIONS AND CAUTION

Important considerations when interpreting MYELIN results.

MYELIN describes software-observed behaviour. It does not directly measure dopamine, neural firing, motivation, boredom, stress, fatigue, or subjective effort.

4 | RESULTS AND DISCUSSION OVERVIEW

Certified behavioural results for experiment PHASE4_TWO_ORGANS in parent session 42.

HOW THIS CHAPTER IS ORGANISED

The overview is followed only by organ pages supported by certified, experiment-scoped quantitative evidence.

SESSION-LEVEL OVERVIEW

Each card states the certified result availability for this immutable experiment snapshot.

SINE	MINE	COSINE	COSECANT	OVERALL EXPERIMENT
AVAILABLE	OUTSIDE_WINDOW	OUTSIDE_WINDOW	AVAILABLE	CERTIFIED
certified factual SINE displays or MIRL return samples are present	no episode in selected experiment	no episode in selected experiment	certified COSECANT break rows are present	Experiment PHASE4_TWO_ORGANS; session 42

AVAILABLE RESULT TYPES

Tables and charts appear only when their specific certified source fields support them.

Behavioural journey map	Episode tables	Stage / escalation tables	Trend graph	Break timing analysis	Unavailable data flag
Chronological view of user behaviour.	List of episodes with key details.	Progression through stages and levels.	Patterns and trends across time.	Timing and duration of breaks.	Shown whenever evidence does not support a value. Never just means the organ was unused.

WHAT APPEARS ON EACH ORGAN PAGE

Organ pages retain the frozen visual family while unsupported fields remain explicitly unavailable.

SINE
- Behavioural journey map
- MIRL table
- SINE stage table
- Trend graph

MINE
- Messaging pathway
- Response options
- Episode table
- Trend graph

COSINE
- OTT pathway
- Break/watch decisions
- Callback summary
- Trend graph

COSECANT
- Study-break route
- Reminder / final-lock levels
- Break timing table
- Return latency

NO_EVIDENCE, UNAVAILABLE, and CROSS_BOUNDARY_OVERLAP are evidence states; they do not by themselves prove organ non-use.

No mixed contained-plus-overlap disclosure is required for this snapshot.

4 | SINE RESULTS AND DISCUSSION

Certified SINE results for experiment PHASE4_TWO_ORGANS.

SINE SESSION SNAPSHOT

Values below come from the immutable certified Phase-2 report package.

Episodes
2
Distinct certified SINE episode identities represented

SINE stages observed
SINE-1, SINE-2
Factual displayed levels

Verified returns
2
Certified successful MIRL return-onset samples

Relapses
0
Certified Chapter 3 relapse metric

Data status
AVAILABLE
Certified SINE results available

BEHAVIOURAL JOURNEY MAP

Factual SINE displays are shown in certified order; same-stage retries remain distinct.

sine-001: SINE-1 → sine-002: SINE-2 → VERIFIED BEHAVIOURAL RETURN

TABLE 1 | MIRL RECOVERY SUMMARY

Successful return-onset samples; latency is opening SINE display to final successful return verification start.

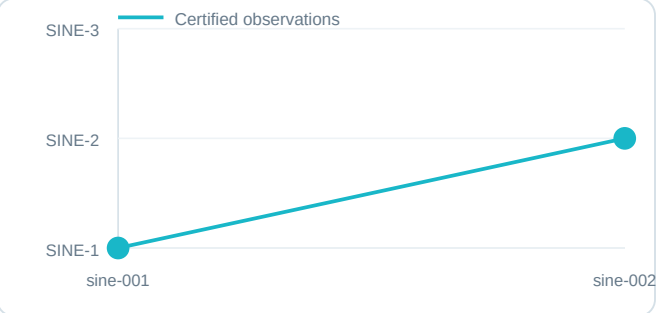
EPISODE	ATTEMPT	FIRST SINE DISPLAY	RETURN ONSET LATENCY
sine-001	attempt-0	2026-09-15T10:00:00Z	2.0 s
sine-002	attempt-1	2026-09-15T10:01:00Z	2.0 s

TABLE 2 | SINE STAGE SUMMARY

Every factual display is retained, including same-stage retries with distinct popup identities.

EPISODE	DISPLAY TIME	SINE LEVEL	RESPONSE
sine-001	2026-09-15T10:00:00Z	SINE-1	UNAVAILABLE
sine-002	2026-09-15T10:01:00Z	SINE-2	UNAVAILABLE

SINE TREND GRAPH



DISCUSSION NOTE

The page reports software-observed SINE displays and MIRL-verified behavioural return. It does not infer cognitive recovery.

4 | COSECANT RESULTS AND DISCUSSION

Certified COSECANT results for experiment PHASE4_TWO_ORGANS.

COSECANT SESSION SNAPSHOT

Values below come from the immutable certified Phase-2 report package.

Approved study breaks
2
Certified COSECANT break identities

Reminder levels observed
1
Factual +5 reminder events

Final locks
0
Factual final-lock events

Return confirmations
2
Factual exact-resource confirmations

Data status
AVAILABLE
Certified COSECANT results available

STUDY-BREAK RETURN PATHWAY

Only certified COSECANT stages actually reached in this experiment are shown.

BREAK STARTED → BREAK COMPLETE → +5 MIN REMINDER → RETURN CONFIRMED

TABLE 1 | BREAK TIMING SUMMARY

Event times are shown as offsets from the certified experiment start; latency is the frozen BREAK_COMPLETE to RETURN_CONFIRMED value.

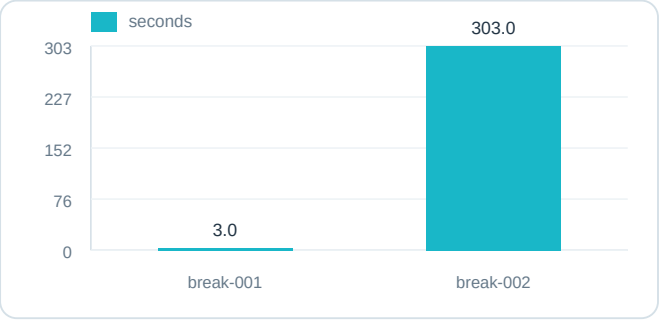
BREAK	BREAK STARTED	BREAK COMPLETE	RETURN LATENCY
break-001	T+00:00:40	T+00:01:40	3.0 s
break-002	T+00:08:40	T+00:09:40	5 min 3.0 s

TABLE 2 | COSECANT ESCALATION SUMMARY

Only factual stages reached by each break are shown.

BREAK	FACTUAL STAGE	EVENT TIME	LATENCY STATUS
break-001	BREAK COMPLETE	T+00:01:40	AVAILABLE
break-001	RETURN CONFIRMED	T+00:01:43	AVAILABLE

COSECANT TREND GRAPH



DISCUSSION NOTE

COSECANT results report exact Study-resource behavioural confirmation. An early return before break completion remains NOT_APPLICABLE, never zero.

4 | COSECANT RESULTS AND DISCUSSION -- CONTINUED

TABLE 2 | COSECANT ESCALATION SUMMARY. Certified rows continue below.

BREAK	FACTUAL STAGE	EVENT TIME	LATENCY STATUS
break-002	BREAK COMPLETE	T+00:09:40	AVAILABLE
break-002	+5 MIN REMINDER	T+00:14:40	AVAILABLE
break-002	RETURN CONFIRMED	T+00:14:43	AVAILABLE

Continuation of SP10; table headers are repeated.

4 | UNAVAILABLE ORGAN RESULTS

The following organs do not receive a quantitative result page for this certified experiment. No value, graph, or table is inferred.

MINE

- State: OUTSIDE_WINDOW
- Reason: no episode in selected experiment

COSINE

- State: OUTSIDE_WINDOW
- Reason: no episode in selected experiment

5 | CONCLUSION

MYELIN approaches digital distraction as a behavioural trajectory rather than only as accumulated screen time.

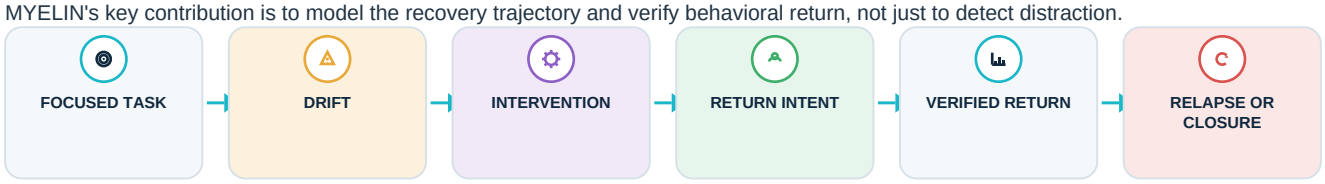
WHAT MYELIN DEMONSTRATES

MYELIN shows that certified digital evidence can provide a structured and meaningful view of distraction and recovery. In this chapter, we conclude that MYELIN:

- Observes the full behavioural trajectory: task engagement, drift, intervention, return, relapse and closure.
- Distinguishes SINE, MINE, COSINE and COSECANT, each with clear meanings and evidence.
- Records certified behavioural evidence from real digital activity.
- Keeps evidence that is unavailable clearly marked as unavailable rather than inferring or inventing data.

RECOVERY AS THE CORE CONTRIBUTION

MYELIN's key contribution is to model the recovery trajectory and verify behavioral return, not just to detect distraction.



MYELIN separates return intention from verified behavioural return. A user's intention to return is recorded, but recovery is only confirmed when subsequent behaviour shows a return to the original task. MIRL is the pathway that carries this recovery evidence -- verification, relapse detection, and closure -- through to a confirmed return.

HCI SIGNIFICANCE

MYELIN provides practical insights for human-computer interaction by showing:

Timing of interventions
How intervention timing relates to observed user responses.

Escalation strength
How different escalation levels relate to observed response patterns.

Burden of prompts
How prompt frequency and escalation relate to observed response patterns.

Supporting self-regulation
Designing for user control and sustainable habits, rather than forcing productivity.

SCIENTIFIC BOUNDARY

It is important to be clear about what MYELIN can and cannot claim.

MYELIN measures software-observed behaviour. It does not directly measure attention, dopamine, neural activity, fatigue, motivation, memory strength or cognitive recovery.

CURRENTLY DEFENSIBLE

- Observed digital behaviour and task sequences
- Timing and duration of drift and return
- User responses to interventions
- Behavioural patterns across contexts and over time

NOT YET ESTABLISHED

- Direct measurement of attention or concentration
- Changes in dopamine or neural activity
- Levels of fatigue or mental effort
- Motivation or intention beyond observed behaviour

MYELIN should be judged by whether its evidence supports reproducible questions about distraction, intervention and recovery.

5 | FUTURE SCOPE AND RESEARCH ROADMAP

This page outlines the next scientific and technical directions for MYELIN. These future directions build on the current certified evidence and aim to further improve understanding, validation and real-world application of MYELIN.

FIVE KEY FUTURE DIRECTIONS

MYELIN can grow in several directions. The following areas represent the main focus for future research and development.

1

CONTROLLED HUMAN VALIDATION

Future studies can compare normal work, conventional focus tools, and MYELIN-assisted regulation, measuring verified returns, relapse, return latency, and task-aligned work.

2

MI -- MYELIN INTELLIGENCE

Mi is a future adaptive intelligence layer above the organs, learning from return and relapse history, time of day, session history and user preferences, but not inventing MYELIN policy.

3

ASK MI + LOCAL MODEL

A future local language model may explain and summarise certified MYELIN evidence, but will not decide whether distraction or recovery occurred.

1 Validation | 2 Mi Intelligence | 3 Local Model | 4 SenseHub / Edge | 5 Longitudinal Research

RESEARCH AND DEVELOPMENT ROADMAP

4

SENSEHUB / MYELIN EDGE

A future physical extension may add phone-presence sensing, study-state display, physical controls, and future experimental physiological synchronization, without claiming this is current evidence.

5

LONGITUDINAL BEHAVIOURAL RESEARCH

MYELIN may study changes across days and weeks, such as return latency, relapse patterns, and stable user-specific patterns.

The next scientific step is controlled validation: determine whether MYELIN-assisted regulation produces measurably different behavioural-recovery outcomes compared with control conditions.

These directions are forward-looking and may evolve based on future research, technical development and real-world findings.

5 | SCIENTIFIC MATURITY AND RESEARCH RESPONSIBILITY

MYELIN should grow as a careful research platform, with clear evidence boundaries, minimum necessary observation, and responsible use of participant data.

1 PRIVACY BY MINIMUM OBSERVATION

MYELIN aims to collect only the behavioural evidence needed for the research question, not the maximum amount of personal data. We focus on task engagement, app usage, timing and other high-level behavioural signals, rather than sensitive content such as message text, file content or full browsing history.

2 LOCAL-FIRST DESIGN

Where possible, MYELIN uses local processing on the participant's device to reduce unnecessary external transmission of data. Local execution can lower privacy risks, but it does not by itself guarantee privacy.

3 RESEARCH RESPONSIBILITY

MYELIN follows a set of core principles to ensure responsible and ethical research practice: informed consent, minimum necessary observation, transparent data use, secure local storage, clear experiment boundaries, and participant control.

Ebbinghaus

- Motivates an elapsed-time / temporal hypothesis about how behaviour changes with time away from task, not a memory-forgetting curve.

JITAI

- Motivates questions on just-in-time, context-aware interventions, but does not prove that MYELIN's timing or delivery is optimal.

Weber-Fechner

- Motivates questions about how changes in stimulus or burden may relate to perceived effect, but does not prove the exact form of the relationship in MYELIN.

Gloria Mark

- Motivates questions about attention, interruption and resumption, but does not prove that MYELIN reduces distraction or improves focus.

CURRENTLY DEFENSIBLE

- MYELIN can collect and organise behavioural data as designed.
- The system can generate descriptive insights about user behaviour (timing, patterns, platform usage).
- The platform provides a structured framework for studying software-observed behaviour in real-world digital work settings.

NOT YET ESTABLISHED

- MYELIN directly improves memory, focus or well-being.
- The observed behavioural changes are caused by MYELIN rather than other factors.
- The results generalise to broader populations and long-term use.

These frameworks motivate testable questions; they do not prove that MYELIN is effective. As MYELIN collects more certified sessions and diverse data, we can address these questions with further analysis, comparisons and longer-term studies.

REFERENCES

Key references cited in this report

This section lists the key academic and industry sources actually cited throughout the MYELIN report. Entries are drawn from the Science Lab bibliography (SCIENCE_LAB/chapter6_view.py:_REFERENCES) and the supplied Chapter One neuroscience literature sweep once verified -- this Phase 3 layout preview does not assert final bibliographic content.

1 [Verified reference pending bibliography reconciliation -- SP1 citation anchor 1]

2 [Verified reference pending bibliography reconciliation -- SP1 citation anchor 2]

3 [Verified reference pending bibliography reconciliation -- SP1 citation anchor 3]

4 Nahum-Shani I, Smith SN, Spring BJ, Collins LM, Witkiewitz K, Tewari A, Murphy SA. Just-in-Time Adaptive Interventions (JITAIs) in Mobile Health. Annals of Behavioral Medicine, 52(6), 446-462, 2018.

5 [Verified reference pending bibliography reconciliation -- SP1 citation anchor 5]

6 Nahum-Shani I, Murphy SA. Just-in-Time Adaptive Interventions: Where Are We Now and What Is Next? Annual Review of Psychology, 77, 679-703, 2026.

7 Murre JMJ, Dros J. Replication and Analysis of Ebbinghaus' Forgetting Curve. PLOS ONE, 10(7), e0120644, 2015.

8 Stevens SS. On the Psychophysical Law. Psychological Review, 64(3), 153-181, 1957.

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10 Mark G, Gudith D, Klocke U. The Cost of Interrupted Work: More Speed and Stress. Proceedings of CHI 2008, 107-110, 2008.

Only sources actually cited in the report are listed here. This Phase 3 preview reuses the live Science Lab bibliography verbatim where already verified; unresolved SP1 anchors are explicitly flagged, not guessed.